COMMERCIAL SPECIFICATION & PLATFORM PROPOSAL

SM48
ENTERPRISE OPERATING SYSTEM

A powerful, integrated educational operating system featuring online MCQ examinations & student paper marking, real-time teacher-student web chat, smart online attendance, high-velocity CRM campaigns, logistics warehouse dispatch, and the autonomous LUMI AI engine.

ACADEMIC & EXAM HUB

Online timed MCQ examination portal, student written paper uploads, evaluator marking rubrics, and grade publication.

LIVE ACADEMIC COLLABORATION

Direct student-to-teacher web chat, smart attendance tracking, and live timetable orchestration.

MULTI-TENANT CRM & CAMPAIGNS

Omnichannel Meta WhatsApp lead ingestion, 4-tier calling suites, automated quota distribution, and coordinator KPI hubs.

LUMI AI INTELLIGENCE SUITE

Autonomous LUMI Caller voice bot, LUMI Chat assistant, AI marketing image generator, and bilingual call audio analytics.

Prepared By: SM48 Systems & Solutions **Product Version:** SM48 Enterprise v5.5 **Date:** August 2026

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> **SM48 Solution Scope & Customization Flexibility**
> SM48 is delivered as an adaptable enterprise platform. Every workflow, user role, calling tier, and interface module documented in this specification can be customized, tailored, or expanded based on direct mutual discussion and specific institutional requirements.

01 Platform Architecture & Multi-Tenant Hierarchy

SM48 is built on an enterprise **Multi-Tenant Relational Architecture** capable of powering multiple educational brands, faculties, and branches on a unified cloud core.

1.1 Entity Relationship Hierarchy

Entity Level	Database Model	Operational Scope & Functionality
1. Business (Tenant)	`Business` / `CrmSettings`	Represents an autonomous educational brand or branch. Holds dedicated Meta WhatsApp Cloud API credentials, Gemini AI keys, custom call ringtones, and payment gateway profiles.
2. Batch (Cohort)	`Batch` / `BatchMonthlyStats`	Target academic year or class cohort (e.g., 2026 A/L, 2027 A/L, O/L). Establishes coordinator KPI targets (Individual: 40%, Group: 50%).
3. Group (Stream)	`Group`	Academic streams (Physical Science, Biological Science, Commerce, Technology, Arts). Configures bundle discount logic and installment structures.
4. Course (Subject)	`Course` / `TuteStock`	Individual academic courses mapped to lecturers, active months, fee structures, examination modules, and physical warehouse inventory.

02 User Hierarchy: Teachers, Students, Staff & Access Control (RBAC)

SM48 enforces zero-trust Role-Based Access Control, providing role-tailored dashboards and permission boundaries across all stakeholders including teachers, students, and administrative staff:

User Role	Stakeholder Category	Permissions & System Capabilities
SYSTEM_ADMIN / SUPER	Executive Administration	Full unrestricted system access: Database management, CRM configuration, staff management, payment overrides, and **Ghost Mode**.
HEAD MANAGER	Executive Operations	Multi-branch oversight, staff KPI monitoring, coordinator management, and financial matrix review.
TEACHER / LECTURER	Academic Faculty	Teacher dashboard, course curriculum management, paper marking & evaluation rubric, student web chat, and class timetable visibility.
CC MANAGER	Class Coordination	Coordinator performance reviews, lead quota distribution, lecturer tute calendar validation, and roster approvals.
COORDINATOR	Class Coordination	Free Seminar CRM, 4-tier Call Campaigns, Class Timetables, Task Center, Web Inquiries, and Student Data Center.
CALLER	Calling Unit	Lead execution dialer, disposition updates, feedback logging, and callback scheduling.
DELIVERY STAFF	Delivery & Logistics	Order packaging, hold issue resolution, waybill printing, courier dispatch, and tute inventory tracking.
FINANCE STAFF	Finance & Accounts	Payment verification, post-pay scheduling, bank statement reconciliation, and financial ledger balancing.
TECHNICAL STAFF	Technical Operations	Studio hall allocation, Zoom meeting provisioning, secure anti-piracy video player streaming, and incident logs.
STUDENT / USER	End User / Learner	Live class streaming, MCQ exams, paper uploads, live teacher chat, delivery tracking, and community discussions.

03 Online MCQ Examination Engine, Paper Upload & Marking Matrix

SM48 provides a complete digital academic assessment suite, allowing students to take timed examinations, submit written papers, and receive graded rubrics directly through the platform.

1. Timed MCQ Examination Engine

Students take timed multiple-choice exams directly in the web portal. Features automated countdown timers, randomized question ordering, instant automated scoring, and detailed answer explanations upon submission.

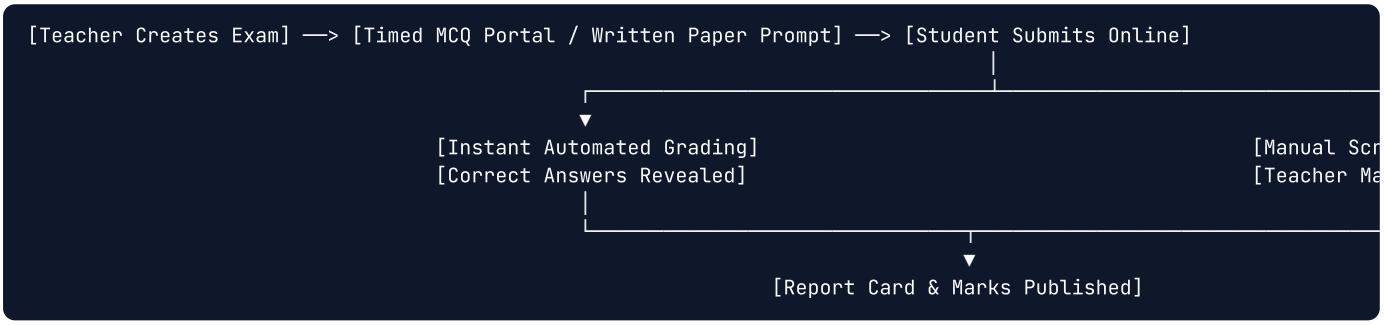
2. Student Paper Upload Hub

For structured essay and written exams, students scan and upload handwritten answer papers as multi-page PDFs or image attachments directly linked to the examination module.

3. Teacher Paper Marking Matrix

Teachers and designated paper evaluators access submitted scripts, annotate corrections, input section marks, award final grades, and publish scores directly into the student's digital report card.

3.1 Examination Lifecycle in SM48



04 Live Student–Teacher Web Chat & Smart Online Attendance

SM48 bridges real-time communication between students and faculty while automating class attendance tracking across online live lectures.

Live Web Chat with Teachers & Coordinators

- Direct one-on-one web chat between students, lecturers, and academic coordinators.
- Rich media support (PDF notes, diagrams, voice clips, and past paper snippets).
- Staff and teachers can respond via the CRM workspace with pre-configured Quick Replies.
- Complete historical chat transcript saved to student profile for continuity.

Smart Online Attendance Tracking

- **Live Join Logging:** Automatically registers student join timestamps and total live lecture watch duration.
- **Participation Verification:** Confirms live presence during active class hours.
- **Attendance Analytics:** Attendance rate reporting per course with automated alerts for irregular attendance.
- **Parent Sync:** Generates automated attendance summary reports accessible by academic management.

05 Class Coordination Department

The Class Coordination Department operates as the daily mission control of the institute, orchestrating timetables, task assignments, rosters, and live student support.

5.1 Departmental Modules & Workflows

1. Coordinator Daily KPI Hub

Real-time progress bars monitoring daily lead engagement against target benchmarks: Individual Goal (40%) and Group Goal (50%), remaining calling quotas, and active classes scheduled for the day.

2. Class Timetable Engine

Comprehensive live schedule manager mapping start/end times, allocated halls, streaming links (Zoom, YouTube, Vimeo), and assigned coordinators with automated pre/in/post-class task generation.

3. Task Center with Proof Validation

Template-driven checklists enforcing operational deadlines. Coordinators complete duties by uploading screenshot proof. Overdue tasks lock automatically, requiring manager approval with written reason to unlock.

4. Lecturer Tute Stock Calendar

Forecasts printed tute requirements against scheduled lecture dates. Cross-references real-time warehouse inventory to trigger low-stock alerts before classes occur.

06 Free Seminar CRM & Inbound Lead Ingestion

The Free Seminar CRM captures, qualifies, and converts prospective seminar attendees into fee-paying enrollments.

6.1 Key Capabilities

- **Meta WhatsApp Cloud API Webhook:** Real-time inbound lead ingestion across multiple WhatsApp Business numbers.
- **Auto Assign Quota Engine:** Fair round-robin distribution bounded by daily coordinator capacity limits.
- **WebRTC WhatsApp Voice Calling:** In-browser calling directly to student WhatsApp numbers with interactive template permissions.
- **Bilingual Call Transcription:** Automatically transcribes voice recordings and analyzes sentiment (Mood, Summary, Vibe Score).

07 After Seminar CRM & The 4 Call Campaign Suites

Following promotional seminars, leads transition into the specialized **4-Tier Call Campaign Suite** to maximize enrollment conversion and reduce student churn.

Campaign Tier	Target Audience	Operational Logic & Progression
1. Open Seminar Leads	Students who attended or registered for promotional seminars.	Multi-round progression (Round 1 → Round 2 → Round 3). Tracks disposition codes (Answered, Busy, Switched Off, Thinking) and calculates delay penalties (`r1DelayedDays`) if follow-ups exceed SLA.
2. New Inquiries	Hot inbound leads from digital marketing and website inquiry forms.	Enforces a high-urgency SLA countdown timer. Requires rapid first-call resolution and follow-up callback scheduling.
3. Bridge Transfers	Leads transferred between academic departments or sister institute brands.	Displays complete previous call notes and transfer justification for seamless context-aware re-engagement.
4. Paid Students (Retention)	Enrolled students on installment plans or with dropped payment history.	Tracks installment due dates (Installments 1, 2, 3), post-pay grace periods, recovers dropped students, and promotes subject bundle upgrades.

08 Delivery & Warehousing Logistics Department

Manages physical printed tute distribution from payment approval through courier waybill generation to final doorstep delivery.

8.1 Delivery Hub Pipeline Tabs

Dashboard Tab	Functional Operations
1. Overview	Executive summary metrics showing orders packed today, pending dispatches, active delivery holds, and completed shipments.
2. Pending & Holds	Isolates orders stalled by incorrect phone numbers, ambiguous addresses, or stock shortages. Features a coordinator collaboration loop to fix and auto-release held orders.
3. Delivered / Dispatched	Tracks shipped parcels with live courier tracking numbers, bulk waybill printing, and delivery date stamping.
4. Delivery History	Complete search archive with instant indexing by student name, telephone number, payment ID, or courier tracking code.
5. Tute Stock & Reports	Live warehouse inventory balance, minimum threshold alerts, auto-deduction upon packing, and historical audit logs.

09 Finance Management, Ledger Balancing & Revenue Matrix

Manages student billing, payment verification, installment schedules, excess/arrears ledger balancing, and bank statement reconciliations.

Payment Verification Hub

Manual review and approval pipeline for student payment receipts supporting Full Payments, Monthly Subscriptions, 3-Tier Installment Plans, and Post-Pay grace periods.

Ledger Balancing (Excess & Arrears)

Automatically records student overpayments or underpayments onto their digital ledger, adjusting next month's fee liability accordingly.

Bank Statement Reconciliation

Imports bank statement CSV feeds to automatically cross-match student claimed payments against verified bank ledger deposits.

Upgrade Pricing Matrix

Calculates exact pro-rated price differentials when students upgrade from single subjects to multi-subject combo bundles.

10 Technical Operations & Anti-Piracy Video Streaming

Coordinates live broadcast infrastructure, studio hall allocations, Zoom account rotation, and anti-piracy video delivery.

10.1 Anti-Piracy Watermarked Video Player

Live streams and recorded lectures are rendered through a proprietary encrypted player with a dynamic floating watermarking layer. The student's full name, telephone number, and IP address animate across the screen at randomized intervals, deterring unauthorized screen recording.

10.2 Zoom Multi-Account Provisioning

Maintains a pool of Zoom OAuth API accounts, automatically provisioning live class meetings, managing concurrent attendee limits, and injecting live access links directly into student dashboards.

11 LUMI AI Suite & Operational Terms

The LUMI AI Suite embeds advanced artificial intelligence across academic, customer service, and marketing workflows.

1. LUMI Chat

24/7 intelligent conversational chatbot responding instantly to student inquiries with institute knowledge base grounding.

2. LUMI Caller

Autonomous AI voice bot making natural outbound seminar reminder and follow-up calls in English & Sinhala.

3. LUMI Image Studio

Generative marketing engine crafting high-resolution promotional banners and graphic assets on demand.

4. AI Call Audio Analytics

Speech-to-text transcription engine that records, transcribes, and calculates sentiment/vibe scores for coordinator calls.

Operational Terms & API Cost Policy for LUMI AI

All LUMI AI capabilities connect directly to foundational LLM and speech inference APIs. **Direct third-party AI API token and computation expenses (e.g., Google Cloud / OpenAI / Anthropic consumption) are borne directly by the purchasing company/client** based on actual institutional usage volume.

12 Student Web Portal & Mobile App Integration Option

12.1 Student Web Portal Features

- **Student Dashboard V2:** Modern web portal highlighting today's scheduled classes, live lecture join links, online MCQ assessments, and campus announcements.
- **Delivery Tracking Hub:** Live multi-stage parcel progress tracker (Order Approved → Packed → Dispatched → Out for Delivery).
- **Paycorp Payment Gateway:** Multi-channel digital fee collection supporting Credit/Debit Cards, QR payments, and manual slip uploads.
- **Moderated Student Community Chat:** Dedicated batch-level discussions featuring automated AI content moderation that removes inappropriate language in real time.

12.2 Mobile Application Add-On Package (LKR 100,000/-)

Native Mobile App Development Option (iOS & Android) — LKR 100K

SM48 is natively delivered as a 100% responsive Progressive Web Platform. If your institute requires a **dedicated Native Mobile Application published on Google Play Store and Apple App Store**, our engineering team will build, brand, and deploy the mobile app for a fixed add-on investment of **LKR 100,000/- (100K)**. All backend endpoints and WebSocket pipelines are already 100% mobile-ready.

13 Commercial Pricing, API Tariffs & Marketing Messaging

The following official pricing and tariff schedules apply to platform channels, telephony, and communication gateways:

1. Meta WhatsApp Cloud API Channel & Marketing Messages

LKR 7,500/- per Meta API Account / Month

Includes unlimited inbound WhatsApp messaging throughput, CRM chat workspace, auto-reply sequence execution, and customer service conversations.

 Meta Outbound Marketing Template Messages: LKR 25.00 per delivered marketing template message (standard Meta API tariff).

2. CRM WhatsApp Voice Caller & Custom SIP Trunk Telephony

FIXED LINE RENTAL	INBOUND CALLS	OUTBOUND CALLS
LKR 2,500/-	**FREE (LKR 0/-)**	**LKR 5.00 / min**
Per Meta API Number / Month	Unlimited Inbound Duration	Per Active Outbound Minute

 Custom Normal Voice SIP Trunk Integration: If your company/institute already possesses or purchases a standard voice SIP Trunk line (e.g. SLT / Dialog / Mobitel SIP), our team will configure and integrate it with the CRM dialer with **ZERO integration fees**.

3. Native Mobile Application Add-On (iOS & Android)

LKR 100,000/- One-Time Development & Store Publishing Package

14 Technical Support, Bug Fixing SLA & Customization Policy

14.1 Continuous Technical Support & Maintenance SLA

We provide full after-sales technical maintenance and operational surveillance:

Bug & Error Resolution

Proactive bug fixing, glitch patching, database query optimization, and performance index tuning included in support coverage.

24/7 System Health Monitoring

Continuous surveillance of WebSocket bridges, Meta webhooks, server memory load, and database response times.

14.2 Future Customization & Extended Development Policy

Bespoke Customization & Future Development Pricing

All features and modules documented in this specification can be customized to your requirements. **Any additional custom development, new departmental modules, or proprietary third-party integrations beyond the scope of this specification will be quoted and delivered upon direct mutual discussion and agreement.**